Bilingual Speech Synthesis with FastSpeech 2

An English and Telugu Study

Technical Report

English (Svarah) and Telugu (IndicTTS Telugu) Neural Text-to-Speech Systems

Project focus
A shared non-autoregressive FastSpeech 2 architecture adapted to English and Telugu through language-specific data preparation, configuration, training, and synthesis.

Prepared as a technical documentation report

1. Overview

This work builds a non-autoregressive neural text-to-speech (TTS) pipeline based on FastSpeech 2 for two languages: English, trained on the Svarah corpus, and Telugu, trained on the IndicTTS Telugu corpus. The codebase follows the open-source PyTorch FastSpeech 2 reference and is adapted to each language through its own corpus, preprocessing rules, and output locations.

The objective is to train a model that learns phoneme timing, fundamental frequency (F0), loudness contours, and the mel-frequency spectrogram in a single feed-forward pass. A HiFi-GAN neural vocoder converts the predicted spectrogram into a waveform. The documented deliverables include per-language configuration files, synthesized audio, prosody-predictor plots, and five evaluation utterances per language assessed for pronunciation and naturalness.

2. System Design and Method

2.1 Base Implementation

The system derives from the xcmyz PyTorch implementation of FastSpeech 2. Its processing chain contains:

- A transformer phoneme encoder that embeds the input phoneme sequence.
- A prosody (variance) predictor that estimates phoneme duration, F0, and loudness.
- A spectrogram decoder that reconstructs the mel-frequency spectrogram, followed by a post-net refinement stage.
- A HiFi-GAN vocoder that renders the waveform from the spectrogram.

The overall inference pipeline is summarized below:

The retained design choices include per-phoneme F0 and loudness prediction, normalized F0 and loudness targets, a six-layer decoder instead of the original paper's four-layer decoder, gradient clipping, learning-rate warm-up, and speaker embeddings for multi-speaker training.

2.2 Retained Architecture Settings

Parameter	Value	Parameter	Value
transformer.encoder_layer	4	transformer.encoder_hidden	256
transformer.decoder_layer	6	transformer.decoder_hidden	256
transformer.encoder_head	2	transformer.decoder_head	2
variance_predictor.filter_size	256	variance_predictor.kernel_size	3
variance_embedding.pitch_quantization	linear	variance_embedding.energy_quantization	linear
vocoder.model	HiFi-GAN	multi_speaker	true

These values agree with the English and Telugu configuration documentation, which emphasizes per-phoneme prosody modelling and a deeper decoder.

2.3 Language-Specific Adaptations

The same network is reused for both languages. The language-specific elements are preprocessing and text-cleaning routines, corpus locations, validation-set size, language tag, lexicon and text-normalization rules, and output folders for audio and prosody plots.

Setting	English	Telugu
language	en	te
text_cleaners	english_cleaners	english_cleaners
validation set size	691	525

Although the architecture is identical, Telugu presents a more challenging modelling task because its script and syllabic structure introduce greater variation in phoneme timing and prosody.

3. Data, Preprocessing, and Training Setup

3.1 English Corpus

The English setup, defined in English/config/English/preprocess.yaml, reads raw data from /kaggle/temp/work/raw_data/English and writes processed data to /kaggle/temp/work/preprocessed_data/English. The configuration uses language tag en, a 22,050 Hz sampling rate, 80 mel channels, normalized phoneme-level F0 and loudness, and 691 held-out validation utterances. Phoneme alignment relies on Montreal Forced Aligner (MFA) with the lexicon lexicon/librispeech-lexicon.txt.

3.2 Telugu Corpus

The Telugu setup, defined in Telugu/config/Telugu/preprocess.yaml, uses /kaggle/temp/work/raw_data/Telugu and /kaggle/temp/work/preprocessed_data/Telugu, language tag te, and a validation set of 525 utterances. The preprocessing logic is unchanged, while the data are expected to be acoustically and phonologically more complex because of vowel–consonant combinations, orthographic variation, and prosodic phrasing.

3.3 Preprocessing Procedure

- Run prepare_align.py to prepare the data for alignment.
- Obtain phoneme-level alignments using MFA or precomputed alignment files.
- Run preprocess.py to extract mel spectrograms, F0, loudness, phoneme durations, and text/phoneme sequences.
- Store the processed results under the corresponding preprocessed_data directory.

Audio / feature setting	Value
Sampling rate	22,050 Hz
STFT filter length / window length	1024 / 1024
Hop length	256
Mel channels	80
F0 and loudness resolution	Phoneme level, normalized

Per-phoneme feature prediction is preferred because the configuration notes report more natural prosody than per-frame prediction.

3.4 Optimization Schedule

Both languages use nearly the same train.yaml configuration:

Hyperparameter	Value	Hyperparameter	Value
batch_size	16	total_step	40,000
grad_clip_thresh	1.0	save_step	10,000
weight_decay	0.0	val_step	1,000
warm_up_step	4,000	anneal_steps	[28,000, 34,000]
anneal_rate	0.3		

The schedule combines a warm-up phase with learning-rate reductions at later stages. Checkpoints and logs are stored in separate directories for each language.

4. Loss-Curve Analysis

Loss plots are stored in Results/English/ and Results/Telugu/, with training_loss.png and validation_loss.png in each language folder.

4.1 Convergence Behaviour

- A sharp drop in loss occurs during the first several thousand steps.
- Improvement becomes slower and steadier during later steps.
- Loss settles into a lower region after warm-up and learning-rate annealing.
- Validation loss follows training loss with a modest gap.

This behaviour indicates stable learning of the mapping from phonemes to spectrogram and prosodic targets. Warm-up and annealing are useful because the model optimizes several coupled objectives: phoneme timing, F0, loudness, and mel reconstruction.

4.2 Instability and Overfitting

- Telugu shows somewhat larger validation-loss swings than English, reflecting the greater difficulty of its phonology and prosody.
- The training–validation gap is present but not large, suggesting generalization without heavy memorization.
- Small late-stage oscillations may occur as the model balances prosody prediction against spectral reconstruction.

Such behaviour is common in multi-objective TTS systems, which must balance audio quality, prosody, and alignment accuracy. Longer training may require additional regularization or extended fine-tuning to reduce remaining instability.

4.3 English versus Telugu

English converges more smoothly because its task and acoustic patterns are more uniform. Telugu is more challenging because of phonetic diversity, script-specific prosodic patterns, complex vowels and syllable structure, and language-specific timing and intonation. Telugu therefore tends toward slower convergence and greater waveform-level instability on difficult sentences.

5. Evaluation of Synthesized Speech

Generated audio for five utterances per language is stored in Results/English/test_generated_audios/ and Results/Telugu/test_generated_audios/. The assessments below reflect the listening observations documented for the report.

5.1 English Utterances

#	Utterance	Assessment
1	Hello, how are you today?	Clear, natural pronunciation and good pacing. Final intonation is slightly flat, but the sample remains smooth and intelligible.
2	The researchers will conduct an experiment.	Fully intelligible with timing that follows phrase structure. Minor roughness occurs around “experiment,” where stress and timing can be compressed.
3	Students learn new things every day.	Highly intelligible with stable prosodic flow and even phrase separation; among the most natural English samples.
4	And they went off into the space when earth was burning up or being destroyed.	The most difficult English item. Long phrasing causes prosodic compression, rushed words, and unstable intonation, although the meaning is retained.
5	Karnataka's expansive list of handmade artisanal products are impressive.	Mostly understandable, but with hesitation-like rhythm and unnatural emphasis. The long, lexically dense sentence reduces smoothness.

5.2 Telugu Utterances

The original Telugu sentences were not recoverable as readable script from the source document. The labels below are therefore retained as placeholders; the original Telugu text should be inserted from the source dataset or audio transcript rather than reconstructed.

#	Utterance label	Assessment
1	[Telugu greeting phrase – insert original]	Clear, understandable greeting with good naturalness in opening and closing tones; pitch variation is strong but acceptable.
2	[Telugu sentence 2 – insert original]	Very intelligible and natural, with stable timing and strong clarity.
3	[Telugu sentence 3 – insert original]	Good overall quality and mostly natural phrase rhythm; some emphasis is slightly exaggerated.
4	[Telugu sentence 4 – insert original]	Generally understandable and expressive. Timing matches clause structure, though sentence edges sound slightly robotic.
5	[Telugu sentence 5 – insert original]	Natural and smooth, with good intelligibility and phrase flow. More stable than the long English sentence.

5.3 Overall Quality

In both languages, the most natural speech occurs on short, syntactically simple sentences. Quality decreases on long, complex, or heavily inflected sentences. Both languages are sensitive to sentence length, while Telugu depends more on phrase-level stability because of its prosodic structure and orthographic complexity.

6. Prosody-Predictor Plots: Timing, F0, and Loudness

Plots for each evaluation utterance are saved under Results//variance_adaptor_plots/ in the duration/, pitch/, and energy/ subfolders. They show how the model assigns timing, F0, and loudness to phonemes or syllables.

6.1 Phoneme Timing (Duration)

In both languages, content words and stressed syllables receive longer durations, function words and short connectors receive shorter durations, and sentence-final words are often lengthened. In English utterance 1, timing increases around “hello,” “today,” and the closing question phrase. In utterance 2, “conduct an experiment” is longer than neighbouring function words, matching natural emphasis. Telugu timing is more variable because prosody depends strongly on word shape and accent placement. The greeting (utterance 1) and utterance 4 show longer durations on content-heavy words while function-like segments remain short.

6.2 Fundamental Frequency (Pitch)

F0 is modelled per phoneme and normalized. Typical patterns include higher F0 at word starts or emphasized syllables, a gradual fall across a phrase unless a new idea or question calls for a rise, and a more marked contour near the end of questions. English utterance 1 rises in the first phrase and falls toward the end, while utterance 2 is flatter and declarative. Telugu plots show stronger local variation: the greeting has a pronounced oscillating contour typical of a question, while utterance 5 progresses more evenly.

6.3 Loudness (Energy)

Loudness acts as a proxy for emphasis, with peaks at stressed words, clause boundaries, and key lexical items.

- English utterance 2 peaks around “conduct” and “experiment,” indicating semantic emphasis.
- English utterance 4 has irregular loudness because the long, dense sentence is difficult for the model to interpret.
- Telugu utterances 3 and 5 show strong peaks on content words, supporting natural emphasis but occasionally overshooting local intensity.

6.4 Interpretation

The plots indicate that the model learns meaningful prosodic structure: longer timing on content words, F0 movement around phrase boundaries, and loudness concentrated on meaningful words. They also reveal limitations: long sentences produce compressed or unstable contours, complex syntax can fragment loudness and F0 patterns, and Telugu prosody remains more variable and less stable than English because of script-specific phonology and lexical complexity.

7. English versus Telugu: Comparison

7.1 English Strengths

English performs better overall. Its corpus and preprocessing are more regular, sentence-level prosody is more uniform, and there are fewer script-driven phonological ambiguities. The reference implementation is also naturally suited to English. As a result, English samples are generally smoother, more natural, and more consistent in rhythm.

7.2 Telugu Strengths

The Telugu model performs well on short, clearly structured sentences. It captures stress and speech rhythm reasonably when the phrase is compact, and short greetings or simple statements can sound natural and clear.

7.3 Limitations and Failure Modes

- Long or syntactically complex sentences can produce unnatural rhythm.
- Pronunciation may degrade on rare lexical patterns or dense phrasing.
- Some words receive unstable F0 or loudness assignments.
- Telugu is more sensitive to morphological and prosodic complexity.
- Output may sound robotic or patchy when timing and F0 are poorly aligned.

These issues are common in small-scale TTS systems.

7.4 Suggested Improvements

- Use a larger and more diverse Telugu dataset.
- Train for longer or fine-tune with prosody-specific loss terms.
- Adopt more robust phoneme lexicons and Telugu text normalization.
- Experiment with different vocoder settings and acoustic-model variants.
- Improve tokenization and normalization for language-specific punctuation and script behaviour.
- Evaluate on more than five representative sentences.
- Add objective metrics such as Mean Opinion Score (MOS), intelligibility, and F0 error alongside subjective listening.

8. Reproducibility

8.1 Dependencies

Dependencies are listed in project/requirements.txt, with per-language files at project/English/requirements.txt and project/Telugu/requirements.txt. The software stack follows the typical FastSpeech 2 setup, including PyTorch and audio and machine-learning libraries.

8.2 Configuration Files

The files preprocess.yaml, model.yaml, and train.yaml under project/English/config/English/ and project/Telugu/config/Telugu/ define dataset paths, feature extraction, F0/loudness normalization, the training schedule, and model architecture.

8.3 Execution Commands

Installation

```
cd project
pip install -r requirements.txt
```

English

```
cd project/English
python prepare_align.py config/English/preprocess.yaml
python preprocess.py config/English/preprocess.yaml
python train.py -p config/English/preprocess.yaml -m config/English/model.yaml -t config/English/train.yaml
python synthesize.py --text "Hello, how are you today?" --restore_step 40000 --mode single -p config/English/preprocess.yaml -m config/English/model.yaml -t config/English/train.yaml
```

Telugu

```
cd project/Telugu
python prepare_align.py config/Telugu/preprocess.yaml
python preprocess.py config/Telugu/preprocess.yaml
python train.py -p config/Telugu/preprocess.yaml -m config/Telugu/model.yaml -t config/Telugu/train.yaml
python synthesize.py --text "" --restore_step 40000 --mode single -p config/Telugu/preprocess.yaml -m config/Telugu/model.yaml -t config/Telugu/train.yaml
```

These commands follow the project README and the per-language configuration layout. Replace the empty Telugu synthesis text with the intended Telugu input sentence when running inference.

9. Conclusion

The project demonstrates that a FastSpeech 2 system can be built for both English and Telugu using one shared architecture with language-specific preprocessing and configuration. English output is more consistent and natural, while Telugu shows promise but is more sensitive to prosodic complexity and longer sentences.

The prosody-predictor plots indicate that timing, F0, and loudness are learned meaningfully. However, the quality gap between short and long sentences suggests that more data, longer training, and improved language-specific normalization could improve intelligibility and naturalness. The repository is organized so that configuration files expose the training and inference parameters clearly. With more diverse data and a longer training schedule, both models could be improved further.

Report note: The Telugu evaluation utterances remain placeholders because the original Telugu text was not legible in the supplied report. No Telugu sentences or quantitative scores have been invented.